



A numerical assembly technique for free vibration of plane frames using a shifted and deflated Newton method

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ABSTRACT

A semi-analytic method, namely the Numerical Assembly Technique (NAT) is extended to compute natural frequencies of arbitrary planar frame structures. The frame structure is systematically defined by a set of nodes, beams, bearings, springs, and external loads and the corresponding boundary and interface conditions are formulated. We consider the Timoshenko-Ehrenfest beam theory and take axial, bending and shear deformations as well as linear and rotational inertia into account. Since we use the exact solutions of the governing differential equations in the NAT, the accuracy of the results is independent from the spatial discretization. However, the resulting eigenvalue problem is non-linear and requires a numerical algorithm to solve. We consider three different representations of exact homogeneous solutions and study the numerical stability of the resulting system matrix. To solve for the eigenvalues we propose a Newton method enhanced with a shifted deflation algorithm. The stability and the efficiency of this approach is illustrated in numerical examples.

1. Introduction

The natural frequencies of a model are decisive factors in dynamic analysis. Especially when studying sensitive structures natural frequencies in the high-frequency range should be taken into account. The approximation of those natural frequencies with finite element methods (FEM) could lack accuracy. At this point we want to emphasize the importance of analytic and semi-analytic methods. For simple problems analytical formulas can be found in various textbooks [1–4]. There, functions to compute natural frequencies and mode shapes are derived for e.g. cantilever beams or single-span beams. There is a variety of analytic and semi-analytic methods also applied to more complicated structures [5]. We want to make references to the most used methods found in literature. One widely used method is the Dynamic Stiffness Method (DSM) which uses shape functions tailored to the governing equations and the assembly technique from the finite element method [6]. Due to its general structure, it can be applied to arbitrary frame structures. Recently, it has been extended to functionally graded beams [7] and axially loaded multi-cracked frames [8] among other extensions such as beams carrying spring-mass systems [9] and axially loaded composite Timoshenko-Ehrenfest beams [10]. To analyze multi-stepped beams with various concentrated masses the Transfer Matrix Method (TMM) is often used. With this approach, a matrix transfers the beam state such as the displacement, the rotation, the internal forces, and the bending moment from one end of a uniform beam segment to the other. Anyway, this method's application is limited to multi-span [11] or multi-stepped beams [12] and can, therefore, not be applied to frame structures. This method has been extended to crack identification of stepped beams with multiple edge cracks and different boundary conditions [13]. Another noteworthy method is the Green's Function Method (GFM), which uses the analytical responses of a uniform beam segment to point loads to include concentrated elements in the beam

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model. It is left to solve for the displacement and the rotations of the beam at each element. Hence, the size of the system matrix depends clearly on the number of concentrated elements. Lately, this method has been used to analyze elastically supported Euler–Bernoulli beams [14], cracked beams with elastic boundary conditions [15] and axially functionally graded and non-uniformed beams [16]. The Generalized Function Approach (GFA) follows a similar idea as the GFM, where the discontinuous response of the beam due to concentrated elements and the changes in material parameters is handled with generalized functions such as Dirac delta functions or Heaviside step functions. Contrarily to the GFM, the size of the system matrix is independent of the number of uniform beam sections. A plane vibration problem can, therefore, be assembled by a 2×2 system matrix. The GFA has been extensively studied. Most recently on beams with various in-span supports and masses [17], on discontinuous layered elastically bonded beams [18], on plane structures with mass–spring sub-systems and rotational joints [19] and on beams including tuned mass dampers with spring inertia effects [20]. The Numerical Assembly Technique (NAT) considers the analytical solution of the homogeneous governing equation. Each analytical solution with the enforced boundary and intermediate conditions holds within a uniform beam segment and occupies one row in the system of equations. Hence, the implementation is straight forward. Numerous research papers concerning the NAT have been published. The most recent publications regarding the NAT are the computation of steady-state harmonic vibrations of viscoelastic Timoshenko-Ehrenfest Beams with fractional derivative damping models [21], the application to multi-stepped Euler–Bernoulli beams under arbitrarily distributed loads carrying any number of concentrated elements [5], the free vibration analysis of continuous bridges under vehicles as ambient excitation forces [22] and the application to cracked Timoshenko-Ehrenfest beams carrying spring-mass systems [23]. However, to the best knowledge of the authors, the NAT is only applied to multi-span and/or multi-stepped beams. In this work, the NAT is extended to plane frame structures based on the Timoshenko-Ehrenfest beam theory with various combinative supports, hinges, and springs. Therefore, we systematically introduce a frame structure as a set of nodes, beams, bearings, springs, and external loads and formulate the corresponding boundary and interface conditions. Writing the system of equations in matrix form allows us to solve for the natural frequencies of the plane structure. This is done using a shifted and deflated Newton method (SDNM). This method is suitable to find distinct solutions of non-linear partial differential equations [24] and has recently gained attention in optimization problems where the maxima of multi-dimensional function gradients need to be computed efficiently [25,26]. More than that, the general function to solve the governing equation for transversal vibrations is improved by using the novel approach introduced in [5]. This increases the accuracy of the results, especially when computing natural frequencies in high-frequency ranges.

2. Mechanical models

2.1. Governing equations for a single beam

When considering vibrations in the two-dimensional space, the longitudinal and transversal deformation of an element are implicated. Let ℓ be the length of the considered beam with constant geometrical and material parameters along its local beam axis. We introduce ρ as the density, E the Young's modulus, G the shear modulus, A the cross-section area, and I the area moment of inertia. With the position $\xi \in [0, \ell]$ and the time t , the differential equations for longitudinal vibrations are given by

$$\rho A \ddot{u}(\xi, t) = N'(\xi, t) + n(\xi, t) \quad \text{and} \quad N(\xi, t) = E A u'(\xi, t). \quad (1)$$

With $u(\xi, t)$ as the axial deformation, $N(\xi, t)$ defining the internal axial force and $n(\xi, t)$ the external axial loading force. Further, $()'$ denotes the derivative with respect to ξ and $()$ the derivative with respect to time. In the present paper we focus on the computation of the natural frequencies of a structure and have therefore not included a model for damping effects. For free vibration analysis (1) simplifies to a homogeneous differential equation with the load force $n(\xi, t) = 0$. In the frequency domain (1) is reformulated to

$$U''(\xi) + \frac{\omega^2}{c^2} U(\xi) = 0 \quad \text{where} \quad c^2 = \frac{E}{\rho} \quad (2)$$

with the frequency ω . The differential equations for transversal vibrations governed by the Timoshenko-Ehrenfest beam theory read as

$$-V'(\xi, t) + \rho A \ddot{w}(\xi, t) = q(\xi, t) \quad \text{and} \quad -M'(\xi, t) + V(\xi, t) - I \rho \ddot{\Theta}(\xi, t) = 0. \quad (3)$$

where the transversal deformation $w(\xi, t)$ is described through the internal shear force $V(\xi, t)$, the internal bending moment $M(\xi, t)$, the rotation angle due to pure bending $\Theta(\xi, t)$ and the external transversal loading force $q(\xi, t)$. For $V(\xi, t)$ and $M(\xi, t)$ the equations

$$V(\xi, t) = \kappa G A \gamma(\xi, t) \quad \text{and} \quad M(\xi, t) = -E I \Theta'(\xi, t) \quad (4)$$

hold, respectively. The Timoshenko-Ehrenfest shear coefficient denoted as κ depends on the cross-section of the beam [27,28] and is considered as $\kappa = 5/6$ (the shear coefficient for a rectangular cross-section) in the remainder of this work. Additionally to the rotation angle due to pure bending we introduce the shear angle $\gamma(\xi, t) \neq 0$. This additional degree of freedom contradicts the Euler–Bernoulli hypothesis that the cross-sections remain even. Thus, the angle $w'(\xi, t)$ between the beam axis and the ξ -axis is a superposition of the angle due to bending and the shear angle and reads as

$$w'(\xi, t) = \gamma(\xi, t) + \Theta(\xi, t). \quad (5)$$

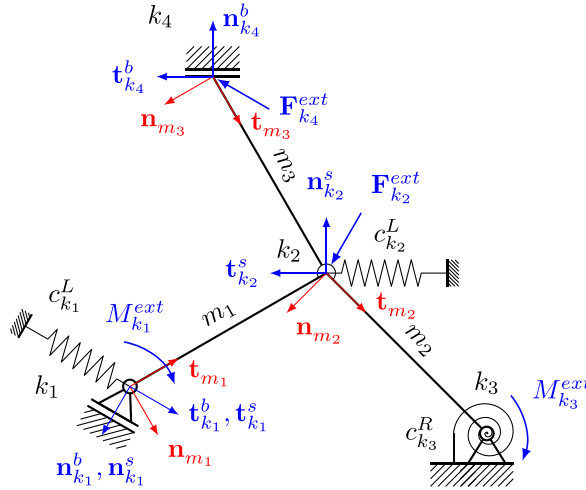


Fig. 1. Sketch of an exemplary planar frame structure. In this case $\mathcal{T} = \{m_1, m_2, m_3\}$ and $\mathcal{N} = \{k_1, k_2, k_3, k_4\}$. Various bearings, springs and external loads are applied. The subsets of connected beams at k_2 read $\beta_{k_2}^h = \{m_3\}$ and $\beta_{k_2}^r = \{m_1, m_2\}$.

The Fourier method of variable separation was applied to find functions satisfying the differential Eqs. (3). The quantities $w(\xi, t)$ and $\Theta(\xi, t)$ lying in the space–time domain are now rewritten as products of the functions $W(\xi)$ and $\psi(\xi)$ dependent on the spatial coordinate ξ and a time-dependent function $T(t)$, such that

$$w(\xi, t) = W(\xi)T(t) \quad \text{and} \quad \Theta(\xi, t) = \psi(\xi)T(t). \quad (6)$$

With the assumptions (6) the system of differential Eqs. (3) can be formulated as

$$W''(\xi) + aW(\xi) - \psi'(\xi) = 0 \quad \text{and} \quad \psi''(\xi) + b\psi(\xi) + cW'(\xi) = 0 \quad (7)$$

in space and

$$\ddot{T}(t) + \omega^2 T(t) = 0 \quad (8)$$

in time, respectively. By eliminating $\psi(\xi)$ from (7) and considering the condition for free vibration $q(\xi, t) = 0$ the single Timoshenko-Ehrenfest beam equations in the frequency domain read as

$$W^{IV}(\xi) + dW''(\xi) + eW(\xi) = 0 \quad \text{and} \quad \psi(\xi) = -\frac{1}{b} [W'''(\xi) + (a+c)W'(\xi)]. \quad (9)$$

where ω is the vibration frequency, and

$$a = \frac{\omega^2 \rho}{\kappa G}, \quad b = \frac{\omega^2 \rho}{E} - c, \quad c = \frac{\kappa G A}{EI}, \quad d = a + b + c, \quad e = ab. \quad (10)$$

2.2. Frame structure

To introduce a frame structure formally, we extend the notation introduced in [29]. Let $\mathcal{N}$ be the set of nodes, $\mathcal{T}$ the set of beams, $\mathcal{B}$ the set of bearings, $\mathcal{S}$ the set of springs and $\mathcal{F}$ the set of external loads. For an exemplary frame structure ($\mathcal{N}, \mathcal{T}, \mathcal{B}, \mathcal{S}, \mathcal{F}$), see Fig. 1, these sets have the following properties:

1. Each beam $m \in \mathcal{T}$ connects two nodes $k_{m,0}, k_{m,\ell} \in \mathcal{N}$ and holds constant parameters $\rho_m, E_m, G_m, I_m, A_m$, the kinetic functions $N_m(\xi), V_m(\xi), M_m(\xi)$ and the displacement functions $U_m(\xi), W_m(\xi), \psi_m(\xi)$ from Section 3. The nodes $k_{m,0}$ and $k_{m,\ell}$ of a beam m can be hinged or rigidly connected to bearings and/or other beams.
2. Each node $k \in \mathcal{N}$ has a set of attached beams $\beta_k \subseteq \mathcal{T} : |\beta_k| \geq 1$ which is the union of two subsets $\beta_k := \beta_k^h \cup \beta_k^r$. Here, in β_k^h are the beams connected by a hinge and in β_k^r are the rigidly connected beams at k . Furthermore, to a node k we associate sets of attached longitudinal springs $\mathcal{S}_k^L \subseteq \mathcal{S}$ and rotational springs $\mathcal{S}_k^R \subseteq \mathcal{S}$, as well as external forces $\mathbf{F}_k^{\text{ext}} \subseteq \mathcal{F}$ and external moments $M_k^{\text{ext}} \in \mathcal{F}$. Finally, a possibly attached bearing $\gamma_k \in \mathcal{B}$ restricts $b_k \leq 3$ degrees of freedom at node k . Those restrictions respectively evoke the constrained force $\mathbf{F}_k^b$ and constrained moment M_k^b .

Here, we assumed that the external forces and moments only act on the nodes. In general, also distributed loads acting on the beams are possible. However, due to the focus on the computation of natural frequencies, the influence of external loads is not further considered in (2) and (9). Further, we assume that rotational springs only act on nodes in the sets $\beta_k : \beta_k^h = \emptyset$ if $|\beta_k| > 1$. To systematically apply kinetic and kinematic boundary and coupling conditions it is crucial to introduce local coordinates for each

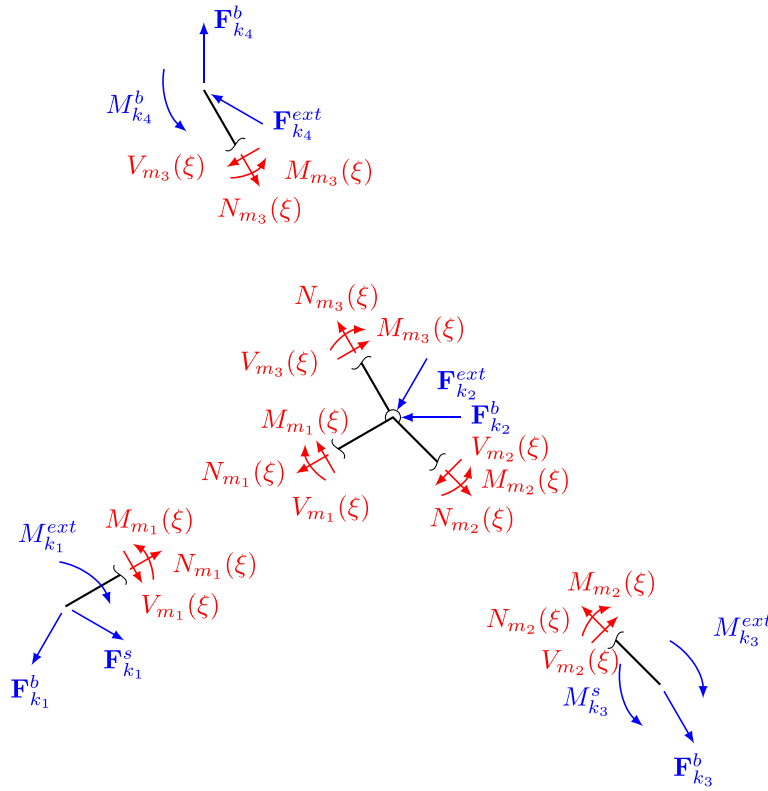


Fig. 2. The free body diagram of the exemplary frame.

beam, longitudinal spring and bearing with a translational degree of freedom. For a beam $m \in \mathcal{T}$ we define the normalized tangent vector $\mathbf{t}_m$ as

$$\mathbf{t}_m = \frac{\mathbf{k}_{m,\ell} - \mathbf{k}_{m,0}}{\|\mathbf{k}_{m,\ell} - \mathbf{k}_{m,0}\|} \quad (11)$$

pointing from $\mathbf{k}_{m,0}$ to $\mathbf{k}_{m,\ell}$ where $\mathbf{k}_{m,0}$ and $\mathbf{k}_{m,\ell}$ are the position vectors of $k_{m,0}$ and $k_{m,\ell}$, respectively. The normalized normal vector $\mathbf{n}_m$ is then defined as the vector $\mathbf{t}_m$ rotated clockwise by 90° . The resulting internal force and deformation function of a beam $m \in \mathcal{T}$ are respectively defined in the local coordinate system $\mathbf{t}_m, \mathbf{n}_m$ as

$$\mathbf{S}_m(\xi) = N_m(\xi) \cdot \mathbf{t}_m + V_m(\xi) \cdot \mathbf{n}_m, \quad \text{and} \quad \mathbf{U}_m(\xi) = U_m(\xi) \cdot \mathbf{t}_m + W_m(\xi) \cdot \mathbf{n}_m. \quad (12)$$

The respective traces of (12) to the end point k read [29]

$$\mathbf{S}_{m|k} := \begin{cases} \mathbf{S}_m(0) & k = k_{m,0}, \\ -\mathbf{S}_m(\ell_m) & k = k_{m,\ell}, \end{cases} \quad \text{and} \quad \mathbf{U}_{m|k} := \begin{cases} \mathbf{U}_m(0) & k = k_{m,0}, \\ \mathbf{U}_m(\ell_m) & k = k_{m,\ell}, \end{cases} \quad (13)$$

and the traces of the internal moment and rotation function are defined respectively as

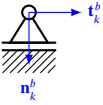
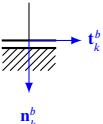
$$M_{m|k} := \begin{cases} M_m(0) & k = k_{m,0}, \\ -M_m(\ell_m) & k = k_{m,\ell}, \end{cases} \quad \text{and} \quad \psi_{m|k} := \begin{cases} \psi_m(0) & k = k_{m,0}, \\ \psi_m(\ell_m) & k = k_{m,\ell}. \end{cases} \quad (14)$$

Due to the usual sign conventions we added the minus for the internal forces and moments in (13) and (14). Similar to the definition of local coordinates of beams we introduce the pair of orthogonal and normalized vectors $\mathbf{t}_k^s, \mathbf{n}_k^s$ as local coordinate systems for longitudinal springs and $\mathbf{t}_k^b, \mathbf{n}_k^b$ for the considered bearings. The quantities $\mathbf{F}_k^s$ and M_k^s are the respective reaction forces and moments due to attached springs from the sets $\mathcal{S}_k^L$ and $\mathcal{S}_k^R$ and are defined through Hooke's law as

$$\mathbf{F}_k^s = -c_k^L (\mathbf{U}_{m|k} \cdot \mathbf{t}_k^s) \cdot \mathbf{t}_k^s \quad \text{and} \quad M_k^s = -c_k^R \psi_{m|k} \quad (15)$$

where c_k^L and c_k^R define the respective spring constants of the longitudinal and the rotational springs. When a bearing γ_k allows one translational degree of freedom we need to consider the constrained force into the direction of $\mathbf{n}_k^b$. For an illustration of the introduced normal and tangent vectors we refer to the example frame in Fig. 1 and the free body diagram in Fig. 2.

Table 1
Kinematic conditions for common bearings at a node k .

Symbol	Name	Condition	Considered set
	pinned	$\mathbf{U}_{m k} = \mathbf{0}$	$\forall m \in \beta_k$
	clamped	$\mathbf{U}_{m k} = \mathbf{0}$ $\psi_{m k} = 0$	$\forall m \in \beta_k$ $\forall m \in \beta_k^r$
	roller	$\mathbf{U}_{m k} \cdot \mathbf{n}_k^b = 0$	$\forall m \in \beta_k$
	parallel guide	$\mathbf{U}_{m k} \cdot \mathbf{n}_k^b = 0$ $\psi_{m k} = 0$	$\forall m \in \beta_k$ $\forall m \in \beta_k^r$

2.3. Boundary and coupling conditions

With the definitions in Section 2.2 the boundary and interface conditions can now be formulated. First, the kinetics are covered by the nodal equilibrium equations

$$\mathbf{F}_k = \sum_{m \in \beta_k} \mathbf{S}_{m|k} + \mathbf{F}_k^b + \mathbf{F}_k^s + \mathbf{F}_k^{ext} = \mathbf{0} \quad (16a)$$

and

$$M_k = \sum_{m \in \beta_k} M_{m|k} + M_k^b + M_k^s + M_k^{ext} = 0. \quad (16b)$$

Next, we formulate the coupling conditions for a node $k \in \mathcal{N} : |\beta_k| > 1$. Therefore, let $m_f \in \beta_k$ and $m_r \in \beta_k^r$ be beams that are fixed in the respective sets. Then the kinematic compatibility at the node k reads

$$\mathbf{U}_{m_f|k} = \mathbf{U}_{m|k} \quad \forall m \in \beta_k \setminus m_f \quad \text{and} \quad \psi_{m_r|k} = \psi_{m|k} \quad \forall m \in \beta_k^r \setminus m_r. \quad (17)$$

Furthermore, let $m_h \in \beta_k^h$ be a beam that is fixed in the set of beams with a hinged connection at this node. Then, the kinetic conditions are

$$M_{m|k} = 0 \quad \forall m \in \beta_k^h \setminus m_h. \quad (18)$$

It is noteworthy that through the equilibrium equation of moments (16a) the condition $M_{m_h|k} = 0$ holds true. Bearings additionally block translation and rotation components of a beam at node k . The kinematic conditions for common bearings are listed in Table 1.

3. Local solutions of the governing equations

The local solution of the differential Eq. (2) in the frequency domain reads

$$U_m(\xi) = C_{m,1} \cos(\mu_m \xi) + C_{m,2} \sin(\mu_m \xi) \quad (19)$$

with $\mu_m = \frac{\omega}{c_m}$ and $c_m^2 = \frac{E_m}{\rho_m}$ and the internal normal force reads

$$N_m(\xi) = E_m A_m U_m'(\xi). \quad (20)$$

Regarding transversal vibrations, solutions to (9) are searched. Formulating a characteristic equation through an exponential ansatz leads to the roots

$$\tilde{z}_{1,m} = \frac{1}{2}(-d_m + \sqrt{A_m}) \quad \tilde{z}_{1,m} \in \mathbb{R} \quad \forall \omega, \quad (21a)$$

$$\tilde{z}_{2,m} = \frac{1}{2}(-d_m - \sqrt{A_m}) \quad \tilde{z}_{2,m} \in \mathbb{R}^- \quad \forall \omega, \quad (21b)$$

where

$$A_m = d_m^2 - 4e_m \quad A_m \in \mathbb{R}^+ \quad \forall \omega. \quad (22)$$

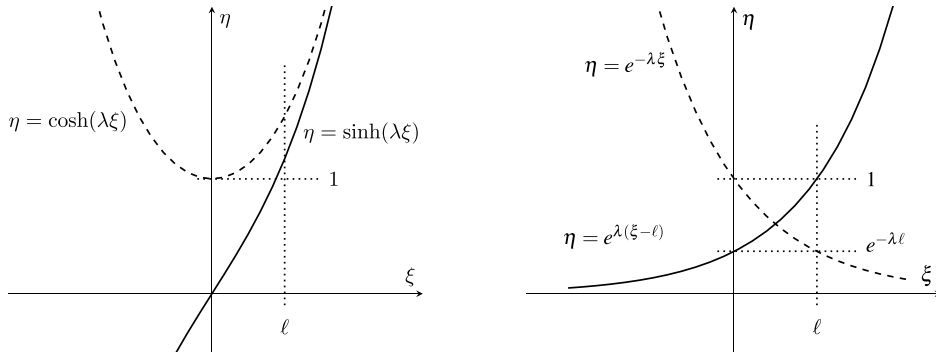


Fig. 3. The difference of the function terms for the general solution.

Clearly there can be a change in signs for $\bar{z}_{1,m}$ forming two spectra of natural frequencies separated by the cut-off frequency $\omega_{co,m}$. The two spectra read as

$$\bar{z}_{1,m} \in \mathbb{R}^+ \forall \omega < \omega_{co,m} \quad \text{and} \quad \bar{z}_{1,m} \in \mathbb{R}^- \forall \omega > \omega_{co,m}. \quad (23)$$

We obtain the cut-off frequency by setting $\bar{z}_{1,m} = 0$ such that

$$d_m = \sqrt{\Delta_m} \Rightarrow \omega_{co,m} = \sqrt{\frac{\kappa G_m A_m}{\rho_m I_m}}. \quad (24)$$

To the best knowledge of the authors, results for $\omega > \min_{m \in \mathcal{I}} \{\omega_{co,m}\}$ are inaccurate and can be neglected [27]. Thus, our focus lies on the first spectrum with $\bar{z}_{1,m} \in \mathbb{R}^+ \forall \omega < \min_{m \in \mathcal{I}} \{\omega_{co,m}\}$ with the general solution

$$W_m(\xi) = P_{m,1} e^{-i\sqrt{\bar{z}_{2,m}}\xi} + P_{m,2} e^{i\sqrt{\bar{z}_{2,m}}\xi} + P_{m,3} e^{-\sqrt{\bar{z}_{1,m}}\xi} + P_{m,4} e^{\sqrt{\bar{z}_{1,m}}\xi}. \quad (25)$$

Many publications [7–14,22,23] refer to the commonly used ansatz

$$W_m(\xi) = \hat{P}_{m,1} \cos(\lambda_{2,m}\xi) + \hat{P}_{m,2} \sin(\lambda_{2,m}\xi) + \hat{P}_{m,3} \cosh(\lambda_{1,m}\xi) + \hat{P}_{m,4} \sinh(\lambda_{1,m}\xi), \quad (26)$$

where $\lambda_{1,m}^2 = |\bar{z}_{1,m}|$ and $\lambda_{2,m}^2 = |\bar{z}_{2,m}|$. Regarding transversal vibrations, in this work, we refer to the improved homogeneous solution. Thus, we apply Euler's formula for trigonometric functions

$$\cos(\xi) + \sin(\xi) = \left(\frac{1}{2} + \frac{i}{2}\right) e^{-i\xi} + \left(\frac{1}{2} - \frac{i}{2}\right) e^{i\xi} \quad (27)$$

to (25), and the equation reads

$$W_m(\xi) = \bar{P}_{m,1} \cos(\lambda_{2,m}\xi) + \bar{P}_{m,2} \sin(\lambda_{2,m}\xi) + \bar{P}_{m,3} e^{-\lambda_{1,m}\xi} + \bar{P}_{m,4} e^{\lambda_{1,m}\xi}. \quad (28)$$

Now, we scale the increasing exponential term $e^{\lambda_{1,m}\xi}$ through dividing it by the constant $e^{\lambda_{1,m}\ell_m}$ such that $e^{\lambda_{1,m}\xi} e^{-\lambda_{1,m}\ell_m} = e^{\lambda_{1,m}(\xi-\ell_m)}$ and the final improved homogeneous solution is

$$W_m(\xi) = P_{m,1}^* \cos(\lambda_{2,m}\xi) + P_{m,2}^* \sin(\lambda_{2,m}\xi) + P_{m,3}^* e^{-\lambda_{1,m}\xi} + P_{m,4}^* e^{\lambda_{1,m}(\xi-\ell_m)}. \quad (29)$$

We excluded the hyperbolic function terms because of their asymptotic form. For higher frequencies or greater beam lengths, the hyperbolic function terms take on high values and make computations unstable and slow. Anyway, increasing exponentials also grow asymptotically and cause instability, hence the scaling of $e^{\lambda_{1,m}\xi}$. Those modifications imply that all function terms are of the form $C\eta_m$ where C is a constant and $\eta_m \in [-1, 1] \forall \omega, \ell_m$ for the trigonometric and $\eta_m \in [e^{-\lambda_{1,m}\ell_m}, 1] \forall \omega, \ell_m$ for the exponential terms, see Fig. 3. These improvements precondition the system matrix by means of the NAT and will be further discussed in Section 4. Finally, the rotation angle of a beam m is

$$\psi_m(\xi) = -\frac{E_m I_m}{\omega^2 \rho_m I_m - \kappa G_m A_m} \left[W_m'''(\xi) + \left(\frac{\omega^2 \rho_m}{\kappa G_m} + \frac{\kappa G_m A_m}{E_m I_m} \right) W_m'(\xi) \right], \quad (30)$$

and the internal shear force and bending moment read respectively as

$$V_m(\xi) = \kappa G_m A_m (W_m'(\xi) - \psi_m(\xi)) \quad \text{and} \quad M_m(\xi) = -E_m I_m \psi_m'(\xi). \quad (31)$$

Remark. In the Dynamic Stiffness Method one uses an alternative (still frequency dependent) basis by enforcing a nodal dirac delta property on the shape functions of the homogeneous solutions $U_m(\xi)$, $W_m(\xi)$ and $W_m'(\xi)$ defined for $\xi \in [k_{m,0}, k_{m,\ell}]$. For longitudinal vibrations such ansatz reads

$$U_m(\xi) = \bar{C}_{m,1} \frac{\sin(\mu_m(\ell_m - \xi))}{\sin(\ell_m \mu_m)} + \bar{C}_{m,2} \frac{\sin(\mu_m \xi)}{\sin(\ell_m \mu_m)}. \quad (32)$$

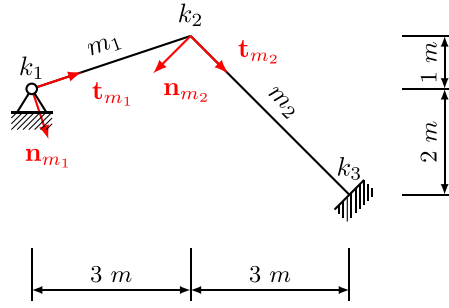


Fig. 4. The sketch of the two-beam frame.

Analogy, for transversal vibrations such an ansatz can be derived. However, the ansatz (32) fails when the considered frequency approaches a clamped-clamped (all nodes of a frame are clamped) natural frequency. To see this, recall the analytic frequency equation for longitudinal vibrations $\sin(\mathcal{L}_m \mu_m) \stackrel{!}{=} 0$. Since the denominators in (32) equal the analytic frequency function for a clamped-clamped beam the ansatz (32) tends to infinity at a clamped-clamped natural frequency and the system matrix becomes ill-posed. The same holds for the transversal vibration ansatz. Anyway, using the NAT with the proposed homogeneous solutions (19) and (29) the matrix entries are well defined for all $\omega \in (0, \infty)$.

4. The numerical assembly technique

We systematically enforce the boundary conditions on the homogeneous solutions (19) and (29) and thus form the system of equations in matrix form as

$$\mathbf{A}(\omega)\mathbf{c} = \mathbf{0} \quad \text{where} \quad \mathbf{A} \in \mathbb{R}^{n \times n} \quad \mathbf{c} := [\mathbf{a}, \mathbf{b}]^T \in \mathbb{R}^{n \times 1} \quad n = |\mathcal{T}| \cdot 6 + \sum_{k \in \mathcal{N}} b_k. \quad (33)$$

The asymmetric system matrix $\mathbf{A}$ depends on ω and the coefficient vector $\mathbf{c}$ contains the vector of integration constants $\mathbf{a}$ and the vector of bearing forces and moments $\mathbf{b}$. Non-trivial solutions for ω and $\mathbf{c}$ fulfill the system of Eqs. (33) and can be computed by finding the roots of the characteristic function, such that $f(\omega) = \det(\mathbf{A}) \stackrel{!}{=} 0$, where $f(\omega)$ is a non-linear function with infinitely many roots. Anyway, calculating the determinant of $\mathbf{A}$ is computationally expensive, therefore, we refer to the shifted and deflated Newton method.

4.1. System matrix assembly of a two-beam frame

An exemplary assembly of $\mathbf{A}$ is given for a simple two-beam frame as depicted in Fig. 4.

To compute discrete matrix entries we project the kinetic and kinematic quantities onto a freely chosen global coordinate system $\mathbf{x}, \mathbf{y}$ with the unit vectors $\mathbf{e}_x \perp \mathbf{e}_y$, and formulate the boundary and coupling conditions, see Table 2, based on the free body diagram Fig. 5. The full matrix is of size 17×17 and can be studied in Appendix.

5. The shifted and deflated Newton method

We are interested in finding the roots of a multidimensional non-linear function $F(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^n$. For our problem, we simultaneously search for the eigenvalues ω and eigenvectors $\mathbf{c}$ in the null space of $\mathbf{A}$ and therefore, refer to the general form of the multidimensional Newton method

$$\mathbf{J}_F(\mathbf{x}_n)(\mathbf{x}_{n+1} - \mathbf{x}_n) = -F(\mathbf{x}_n) \quad (34)$$

where $\mathbf{J}_F(\mathbf{x})$ is the Jacobian of $F(\mathbf{x})$. We want to find $\omega \in \mathbb{R}$ and $\mathbf{c} \in \mathbb{R}^n$ such that

$$\mathbf{r}_1 := \mathbf{A}(\omega)\mathbf{c} = \mathbf{0} \quad \text{and} \quad r_2 := \|\mathbf{c}\| - 1 = 0. \quad (35)$$

The condition $\mathbf{r}_1 = \mathbf{0}$ uniquely defines the eigenvalues of $\mathbf{A}$. However, the eigenvectors are only defined up to a multiplicative constant, i.e. if $\mathbf{c}^*$ is an eigenvector to the eigenfrequency ω^* , then for $a \in \mathbb{R}$, the vector $a\mathbf{c}^*$ is also an eigenvector. In order to fix the multiplicative constant we introduce the condition $r_2 = 0$. This means that we have fixed the Euclidean norm of $\mathbf{c}$ to be 1. Thus, the function reads $F(\omega, \mathbf{c}) = [\mathbf{r}_1, r_2]^T$ and the Newton method (34) can be rewritten in a recursive matrix form as

$$\begin{bmatrix} \mathbf{A}(\omega_i) & \frac{\partial \mathbf{A}(\omega_i)}{\partial \omega} \mathbf{c} \\ \frac{\partial \|\mathbf{c}_i\|}{\partial \mathbf{c}} & 0 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{c} \\ \Delta \omega \end{bmatrix} = - \begin{bmatrix} \mathbf{r}_1(\omega_i, \mathbf{c}_i) \\ r_2(\mathbf{c}_i) \end{bmatrix} \quad (36)$$

Table 2

The boundary and coupling conditions for the two-beam frame.

Node	Row in A	Condition	
k_1	1	$M_{m_1}(0) = 0$	} equilibrium equations
k_1	2	$(\mathbf{F}_{k_1}^b + \mathbf{S}_{m_1}(0)) \cdot \mathbf{e}_x = 0$	
k_1	3	$(\mathbf{F}_{k_1}^b + \mathbf{S}_{m_1}(0)) \cdot \mathbf{e}_y = 0$	
k_2	4	$M_{m_1}(\sqrt{10}) + M_{m_2}(0) = 0$	
k_2	5	$(\mathbf{S}_{m_1}(\sqrt{10}) + \mathbf{S}_{m_2}(0)) \cdot \mathbf{e}_x = 0$	
k_2	6	$(\mathbf{S}_{m_1}(\sqrt{10}) + \mathbf{S}_{m_2}(0)) \cdot \mathbf{e}_y = 0$	
k_3	7	$M_{m_2}(3\sqrt{2}) + M_{k_3}^b = 0$	
k_3	8	$(\mathbf{F}_{k_3}^b + \mathbf{S}_{m_2}(3\sqrt{2})) \cdot \mathbf{e}_x = 0$	
k_3	9	$(\mathbf{F}_{k_3}^b + \mathbf{S}_{m_2}(3\sqrt{2})) \cdot \mathbf{e}_y = 0$	} bearing conditions
k_1	10	$\mathbf{U}_{m_1}(0) \cdot \mathbf{e}_x = 0$	
k_1	11	$\mathbf{U}_{m_1}(0) \cdot \mathbf{e}_y = 0$	
k_3	12	$\mathbf{U}_{m_2}(3\sqrt{2}) \cdot \mathbf{e}_x = 0$	
k_3	13	$\mathbf{U}_{m_2}(3\sqrt{2}) \cdot \mathbf{e}_y = 0$	
k_3	14	$\psi_{m_2}(3\sqrt{2}) = 0$	} coupling conditions
k_2	15	$(\mathbf{U}_{m_1}(\sqrt{10}) - \mathbf{U}_{m_2}(0)) \cdot \mathbf{e}_x = 0$	
k_2	16	$(\mathbf{U}_{m_1}(\sqrt{10}) - \mathbf{U}_{m_2}(0)) \cdot \mathbf{e}_y = 0$	
k_2	17	$\psi_{m_1}(\sqrt{10}) - \psi_{m_2}(0) = 0$	

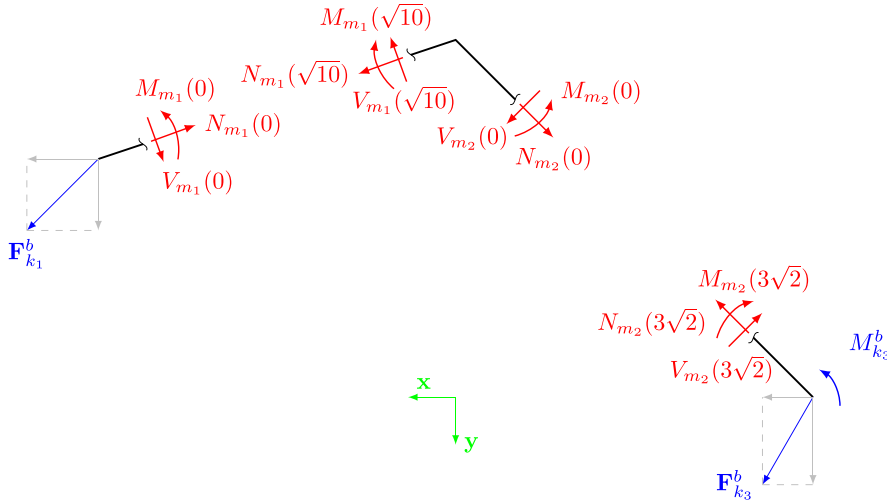


Fig. 5. The free body diagram of the two-beam frame.

where

$$\mathbf{c}_{i+1} = \mathbf{c}_i + \Delta \mathbf{c} \quad \text{and} \quad \omega_{i+1} = \omega_i + \Delta \omega \quad (37)$$

are the iteratively computed solutions of $\mathbf{c}$ and ω , respectively. To make sure no root of $F(\omega, \mathbf{c})$ is computed twice, we introduce the deflation method and iteratively remove the already found roots from the function by rewriting $\mathbf{r}_1$ as

$$\hat{\mathbf{r}}_1 := \mathbf{A}(\omega) \mathbf{c} \prod_{i=1}^N (\omega - \omega_i)^{-2} \quad (38)$$

with the already found roots $\omega_1, \dots, \omega_N$. To adjust the Jacobian we take the partial derivatives of the deflated function $F(\omega, \mathbf{c}) = [\hat{\mathbf{r}}_1, \mathbf{r}_2]^T$. Here, we only mention the rather intricate derivative

$$\frac{\partial \hat{\mathbf{r}}_1}{\partial \omega} = \mathbf{A}(\omega) \mathbf{c} \sum_{j=1}^N -2(\omega - \omega_j)^{-3} \prod_{\substack{i=1 \\ j \neq i}}^N (\omega - \omega_i)^{-2} + \frac{\partial \mathbf{A}(\omega)}{\partial \omega} \mathbf{c} \prod_{i=1}^N (\omega - \omega_i)^{-2}. \quad (39)$$

Regarding (39) the partial derivative can get very small when the distance between the iterative and the already found eigenvalues is big. In the limit case it reads

$$\lim_{|\omega - \omega_i| \rightarrow \infty} \frac{\partial \hat{\mathbf{r}}_1}{\partial \omega} = \mathbf{0} \quad (40)$$

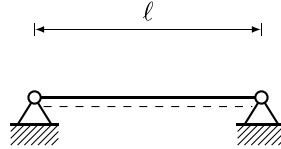


Fig. 6. A sketch of the one-span beam.

and it implies that the computations become inaccurate. We avoid this problem by shifting $\hat{\mathbf{r}}_1$ such that

$$\hat{\mathbf{r}}_1^* := \mathbf{A}(\omega) \mathbf{c} \prod_{i=1}^N [(\omega - \omega_i)^{-2} + 1]. \quad (41)$$

The shifted partial derivative then reads

$$\frac{\partial \hat{\mathbf{r}}_1^*}{\partial \omega} = \mathbf{A}(\omega) \mathbf{c} \sum_{j=1}^N -2(\omega - \omega_j)^{-3} \prod_{\substack{i=1 \\ j \neq i}}^N [(\omega - \omega_i)^{-2} + 1] + \frac{\partial \mathbf{A}(\omega)}{\partial \omega} \mathbf{c} \prod_{i=1}^N [(\omega - \omega_i)^{-2} + 1] \quad (42)$$

and, thus, in the limit case it holds

$$\lim_{|\omega - \omega_i| \rightarrow \infty} (\omega - \omega_i)^{-n} + 1 = 1 \Rightarrow \lim_{|\omega - \omega_i| \rightarrow \infty} \frac{\partial \hat{\mathbf{r}}_1^*}{\partial \omega} = \frac{\partial \mathbf{A}(\omega)}{\partial \omega} \mathbf{c}. \quad (43)$$

Based on our observations we state that with the shifted and deflated Newton method also large systems of equations can be solved rapidly and the time for solving does not heavily depend on the size of the system matrix.

6. Numerical examples

We show improvements through the proposed method by means of three numerical examples. The first example is a one-span beam and primarily serves to investigate the effect of the improved homogeneous solution on the condition of the system matrix. In the second example we study the influence of two different numerical precision types on the proposed method with the three different homogeneous solutions applied at a two-beam frame. Also, the accuracy of finite element computations (Abaqus) based on a global h -refinement is analyzed and compared to the analytic solution. In the last example we investigate the capability of the proposed method using the improved homogeneous solution applied at frames with an increasing number of beams and study the accuracy of finite element computations with different mesh sizes. Finally, we examine the proposed non-linear solver and compare it to the Regular-Falsi method and finite element computations in terms of computational effort.

6.1. One-span beam

We test a one-span beam with the parameters $\ell = 1 \text{ m}$, $A = 7.56 \times 10^{-4} \text{ m}^2$, $I = 3.5 \times 10^{-10} \text{ m}^4$, $\rho = 7.85 \text{ t m}^{-3}$, $E = 2 \times 10^8 \text{ kN m}^{-2}$, and $G = 8 \times 10^7 \text{ kN m}^{-2}$, see Fig. 6.

Investigated is the condition number for the bending terms of the system matrix $\mathbf{A}$ in the range of its first spectrum of natural frequencies $\omega \in (0, \omega_{co})$. It clearly shows in Fig. 7 that the system matrix based on the homogeneous solutions (26) and (28), become ill-conditioned for higher frequencies which leads to an unstable eigenvalue problem. We observe an improvement when using (28) compared to (26). This improvement is obtained by avoiding the hyperbolic terms in the general solution. Anyway, by scaling the exponential term of the homogeneous solution (28), thus using (29) the condition number remains low, even for high frequency ranges. This indicates a well conditioned system matrix and therefore, a stable system of equations. The accurate evaluation of natural frequencies should therefore be improved by the use of (29), particularly for higher frequency ranges.

6.2. Two-beam frame

The considered frame structure consists of two beams and is pinned at node k_1 and clamped at node k_3 . The beams are rigidly connected at node k_2 , see Fig. 8. The frame was modeled based on a steel profile with the parameters $A_m = 7.56 \times 10^{-4} \text{ m}^2$, $I_m = 3.5 \times 10^{-10} \text{ m}^4$, $\rho_m = 7.85 \text{ t m}^{-3}$, $E_m = 2 \times 10^8 \text{ kN m}^{-2}$ and $G_m = 0.8 \times 10^8 \text{ kN m}^{-2}$ for each beam. In Fig. 9 the condition number of the system matrix was plotted, now for transversal and longitudinal terms. We observe a similar behavior at high frequencies as seen for the one-span beam, although the difference between (26) and (28) is more visible when the bending terms are observed only. In Table 3 we show the number of computable natural frequencies using the NAT before the system matrix becomes too ill-conditioned and computations fail. Opposed are the three homogeneous solutions with the precision types *double* and *vpa*. When using *vpa* we expect better results compared to *double* as data with a higher precision has a lower impact on a possibly unstable system matrix. The stability obtained when using the improved homogeneous solution makes it possible to compute all natural frequencies up to the cut-off frequency accurately, even with double precision, see Table 3. The results from Table 3 can be compared with the results from finite element computations in Table 4 where n_{el} is the number of elements per beam, n_t is the total number of computed

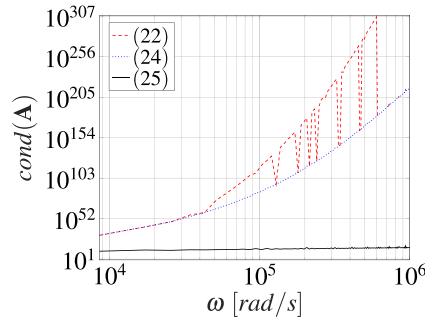


Fig. 7. The condition number of the system matrix.

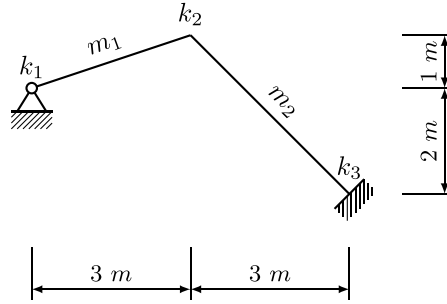


Fig. 8. The modeled two-beam frame.

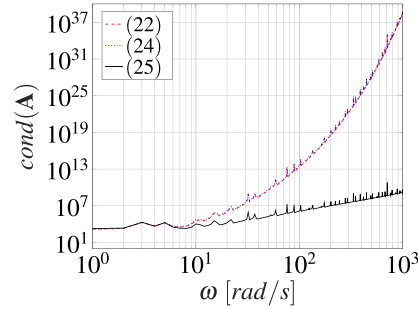


Fig. 9. The condition number of the system matrix.

Table 3
Number of computable natural frequencies using the NAT.

hom. solution	double	vpa
(26)	15	37
(28)	429	765
(29)	All $\omega \in (0, \omega_{co})$	

natural frequencies and $n_{r \leq 10\%}$ is the number of computed natural frequencies with an absolute error $r = \left| \frac{\omega_{FEM} - \omega_{NAT}}{\omega_{NAT}} \right| \leq \frac{1}{10}$ to the analytic solution ω_{NAT} . Table 4 shows that, when using finite elements for the modal analysis, a fine discretization of the structure is required to investigate high frequencies. From Table 4 we can also study that the number of natural frequencies that lie within an acceptable error range highly depends on the mesh size and is generally a small fraction of the total number of computed natural frequencies. For this example $\omega_{co} = 4.2830 \times 10^6 \text{ rad s}^{-1}$ is the 5990th natural frequency.

6.3. Multi-storey frame

With this example we want to show that the NAT is also applicable to more complex frame structures, i.e. structures with multiple connected beams. A building with multiple storeys is a common structure and therefore, serves as our object of investigation. The

Table 4
Number of computable natural frequencies using the FEM.

n_{el}	n_f	$n_{r \leq 10\%}$
16	350	25
32	702	67
64	1406	131

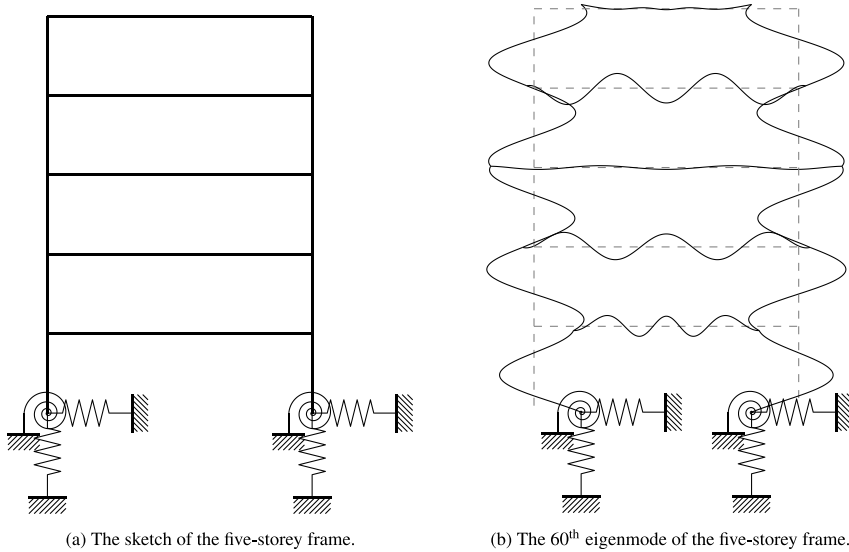


Fig. 10. A plot of an exemplary multi-storey frame with five storeys and a plot of its 60th eigenmode.

Table 5
Computational effort [s] using the SDNM, the RFM and the FEM.

$\#\omega \in (0, \omega_{co})$	#storeys	SDNM	RFM	FEM
79	1	1.1817	0.60673	1.2
155	2	2.5398	5.7607	1.9
231	3	4.7625	10.9671	2.3
382	5	11.6772	25.2154	4.2
532	7	25.2426	–	7.5

dimension $height \times width$ of one storey is $3 \text{ m} \times 10 \text{ m}$, and is shown in Fig. 10(a). The frame is modeled of reinforced concrete with the parameters, $A_m = 8 \times 10^{-2} \text{ m}^2$, $I_m = 11 \times 10^{-4} \text{ m}^4$, $\rho_m = 2.5 \text{ t m}^{-3}$, $E_m = 31 \times 10^6 \text{ kN m}^{-2}$ and $\nu_m = 0.2$ (Poisson's ratio). The shear modulus is calculated as $G_m = E_m(2(1 + \nu_m))^{-1}$. The foundations are elastically embedded in the ground with the spring stiffness $c^L = 10^3 \text{ kN m}^{-1}$ for the longitudinal and $c^R = 10^3 \text{ kNm rad}^{-1}$ for the rotational springs, respectively. We test the presented NAT with the improved homogeneous solution on frames with various storeys. The proposed method solves the non-linear eigenvalue problem by the shifted and deflated Newton method (SDNM). Anyway, we also study the solution by computing the roots of the frequency function $f(\omega) = \det(\mathbf{A})$ by means of the Regular-Falsi method (RFM). The computational efforts of the SDNM and the RFM were compared to the computational costs using the finite element method on the finest considered mesh, see Table 5. The comparison is based on the computation of all natural frequencies within the first frequency spectrum. In Table 6 we study the accuracy of the finite element computations based on a global h -refinement applied to various frames considering the results from the NAT as the exact solution. In Figs. 11 to 15 we plotted all natural frequencies $\omega \in (0, \omega_{co})$ for the considered frames and refinement levels. For these frames $\omega_{co} = 17695.5396 \text{ rad s}^{-1}$. The number of natural frequencies lying within the first frequency spectrum $\omega \in (0, \omega_{co})$ depends on the complexity of the frame and is listed in the tables below. Finally, an exemplary plot of the 60th eigenmode of the five-storey frame is given in Fig. 10(b).

Computational costs. The computational costs for the NAT are slightly higher but comparable to the finite element method. Most of the computational time is spent on solving the linear system of equations within the Newton method. Nevertheless, an improvement is noticeable in Table 5 when comparing the proposed shifted and deflated Newton method to the Regular-falsi method. Also, with the SDNM it is possible to compute the natural frequencies of structures with a higher number of beams. The use of the RFM is limited to a certain complexity of the structure as the frequency function takes on high numbers that exceed the range of normal computers. In this example the natural frequencies of the seven-storey frame cannot be computed with the RFM anymore.

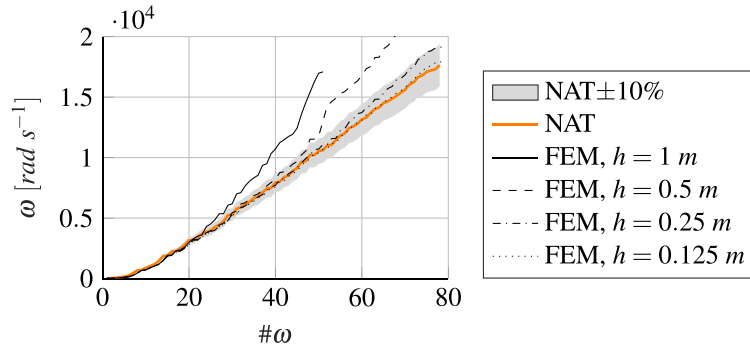


Fig. 11. The natural frequencies of the one-storey frame.

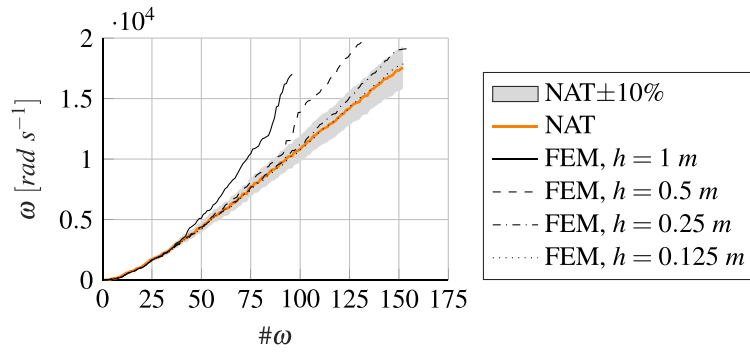


Fig. 12. The natural frequencies of the two-storey frame.

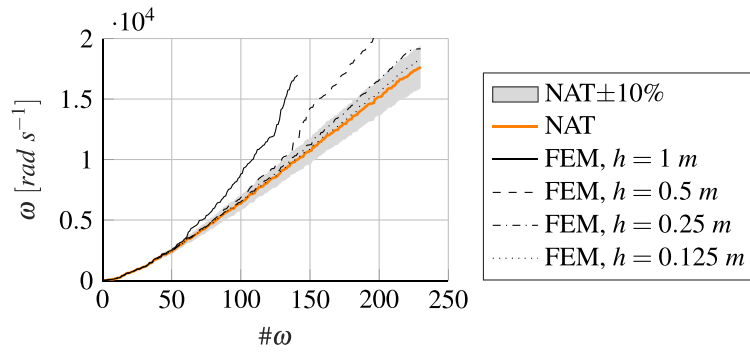


Fig. 13. The natural frequencies of the three-storey frame.

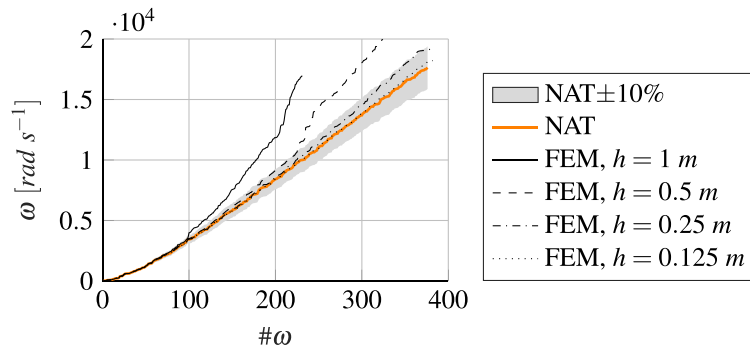
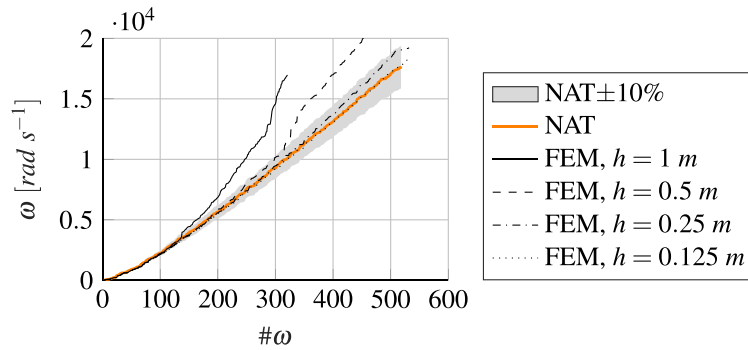


Fig. 14. The natural frequencies of the five-storey frame.

Table 6

Study of the natural frequencies using the FEM with a decreasing element size h . The number inside the parenthesis is the total number of computed natural frequencies. The number outside the parenthesis is the number of computed natural frequencies with an absolute error $< 10\%$.

$\#\omega \in (0, \omega_{co})$	#storeys	$h = 1 \text{ m}$	$h = 0.5 \text{ m}$	$h = 0.25 \text{ m}$	$h = 0.125 \text{ m}$
79	1	22 (51)	43 (99)	72 (195)	79 (387)
155	2	42 (96)	82 (192)	142 (384)	155 (768)
231	3	61 (141)	121 (285)	208 (573)	231 (1149)
382	5	99 (231)	193 (471)	348 (951)	382 (1911)
532	7	109 (321)	283 (657)	532 (1329)	532 (2673)

**Fig. 15.** The natural frequencies of the seven-storey frame.

Accuracy of the results. The main advantage of the proposed NAT is the accuracy of the results and its independence from the discretization of the problem. The Table 6 shows that the mesh size has to be adapted to the wished frequency range and a wished accuracy of the results when using the finite element method. One can estimate a suitable mesh size such that a certain number of elements covers one wavelength. Anyway, when the exact solution is unknown it is difficult to make the right guess. The Figs. 11 to 15 visualize the results from Table 6 and give an idea about the accuracy of the finite element method in modal analysis.

7. Conclusion

By means of the present work, the Numerical Assembly Technique can be used to compute the natural frequencies of planar frame structures. When modeling a structure the user can choose between the common bearings, rotational and longitudinal springs, and is free to define the coupling properties between beams. The Timoshenko-Ehrenfest beam theory is implemented for transversal vibrations. Thus, shear deformation and rotatory inertia is considered. Since analytic solutions are used, the discretization at the nodes is sufficient and exact results are obtained. The improved homogeneous solution ensures stability and efficiency, especially for high-frequency ranges. A shifted and deflated Newton method provides a stable and fast computation of the eigenvalues and eigenvectors of the system matrix.

Enhancements of this work are feasible. Remaining in the two-dimensional setting, one can consider attached concentrated masses and dampers [5], material damping [21], and locally cracked beams [23]. If damping is introduced one deals with a complex-valued system matrix and needs to reassess the shifted and deflated Newton method. A further interesting research direction would be the extension to space frames. This would mean that the boundary and coupling conditions need to be considered for the six degrees of freedom in space and the analytical solutions need to be respectively defined for biaxial bending, torsional and axial vibrations. Researchers have worked on space frames such as [30,31]. In both cases the general solution to torsional vibrations were derived respecting the Saint-Venant torsion theory. This is a differential equation of second order and requires boundary conditions for the torsional deformation or torsional moment. Thus, the implementation is straight forward but limits the application to beams with circular cross sections. Aside from those simplifications, one includes warping effects and the coupled differential equation for flexural-torsional vibrations needs to be considered as in e.g. [32]. In this case, the user also needs to specify additional geometric parameters such as the bending moment of inertia for the out-of-plane bending, the rotatory moment of inertia and the orientation of the cross section in space. As a last further possible research direction we mention the study of forced vibrations. Since the eigenfunctions of the system matrix form an orthogonal basis of the solution space, one might think of computing the forced vibration response of a frame by an eigenfunction expansion of the forcing vector.

CRedit authorship contribution statement

Thomas Kramer: Writing – original draft. **Michael Helmut Gfrerer:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix. The full a matrix of the two-beam frame

The following abbreviations

$$\begin{array}{lll}
K_m = \kappa G_m A_m & B_m = E_m I_m & D_m = E_m A_m \mu_m \\
\Xi_{i,m}^{\pm\Omega} = e^{\pm\Omega\lambda_{i,m}} & \gamma_{i,m}^{\pm} = \frac{\lambda_{i,m}}{b_m} (\pm\lambda_{i,m}^2 + a_m + b_m + c_m) & \delta_{i,m}^{\pm} = \frac{\lambda_{i,m}}{b_m} (\pm\lambda_{i,m}^2 + a_m + c_m) \\
\varepsilon_{i,m}^{\pm} = \frac{\lambda_{i,m}}{b_m} (\pm\lambda_{i,m}^4 + \lambda_{i,m}^2 (a_m + c_m)) & \zeta_{i,m}^{\pm} = \frac{1}{b_m} (\pm\lambda_{i,m} (a_m + c_m) + \lambda_{i,m}^3) & A_{i,m}^{s,\Omega} = \sin \Omega \lambda_{i,m} \\
A_{i,m}^{c,\Omega} = \cos \Omega \lambda_{i,m} & \Upsilon_m^{s,\Omega} = \sin \Omega \mu_m & \Upsilon_m^{c,\Omega} = \cos \Omega \mu_m \\
\tilde{A}_{i,m}^{d,\Omega} = \Gamma_{i,m} A_{i,m}^{d,\Omega} & \Xi_{i,m}^{d,\Omega} = \Gamma_{i,m} \Xi_{i,m}^{d,\Omega} & \mathbf{F}_{k,x}^b = \mathbf{F}_k^b \cdot \mathbf{e}_x \\
\mathbf{F}_{k,y}^b = \mathbf{F}_k^b \cdot \mathbf{e}_y & &
\end{array}$$

for $d = \{s, c\}$ and $\Gamma_{i,m} = \{\lambda_{i,m}, \gamma_{i,m}^{\pm}, \delta_{i,m}^{\pm}\}$ allow us to write the system of equations of the two-beam frame in Fig. 8 as

[illegible]

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